BRIGHTLIGHT TUTORIALS
- DATA Analytics

SQL Fundamentals

SQL EXERCISE 4 : JOINS, UNIONS, Filtering, and Aggregates

SQL EXERCISE: JOINS, UNION, FILTERING AND AGGREGATE

TABLE 1: Employees

Employee ID	First Name	LastName ~~Department~~	Department ~~Salary~~	Salary	City	Age
1	John	Doe	IT	70000	New York	35
2	Alice	Smith	HR	60000	Los Angeles	40
3	Bob	Johnson	Finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	Finance	80000	New York	50

TABLE 2: Projects

Project_Id	Employee ID	Project_Name	Budget	Start Date	Status
101	1	AI Development	100000	2024-01-10	Completed
102	2	Employee Training	50000	2024-02-15	Ongoing
103	1	Cybersecurity Audit	75000	2024-03-05	Pending
104	3	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	80000	2024-06-10	Pending

1.

```
SELECT e.EmployeeID; e.FirstName, e.LastName,
       e.Department, e.Salary, p.ProjectID;
       p.ProjectName, p.Budget, p.Status
FROM Employees
INNER JOIN Projects p.ON e.EmployeeID =
       p.EmployeeID;
```

~~EmployeeID FirstName LastName Department Salary ProjectID ProjectName Budget~~

Outcome:

Employee-ID	First-name	Last-name	Department	Salary	Project-ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Develop	100 000	Completed
1	John	Doe	IT	70 000	103	Cybersecurity	75000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	60000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

1.2.

```
SELECT e.EmployeeID, e.FirstName, e.LastName,
       e.Department, e.Salary, p.ProjectID,
       p.ProjectName, p.Budget, p.Status
FROM Employees
LEFT JOIN Projects p ON e.EmployeeID = p.EmployeeID;
```

Employee ID	FirstName	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

~~Empl ID First Name LastN ame Department Salary Project ID~~

Outcome:

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

3.

```
SELECT p.ProjectID, p.ProjectName,
       p.Budget, p.status, employeeID,
       e.FirstName, e.LastName, e.department
       e.salary
FROM Employee e
RIGHT JOIN Projects p ON e.EmployeeID=
       p.EmployeeID;
```

Employees - Outcome

ProjectID	ProjectName	Budget	Status	EmployeeID	First name	Last name	Department	Salary
101		100000	Completed	1	John	Doe	IT	70000
102		50000	Ongoing	2	Alice	Smith	HR	60000
103		75000	Pending	1	John	Doe	IT	70000
104		90000	Ongoing	3	Bob	Johnson	Finance	75000
105		65000	Completed	5	Emma	Wilson	Sales	65000
106		80000	Pending	6	Michael	Clark	Finance	80000

#4

FULL OUTER JOIN

```
SELECT EmployeeID, FirstName, LastName,
       Department, Salary, ProjectID, ProjectName,
       Budget, Status
FROM Employees
FULL OUTER JOIN Projects ON EmployeeID =
                EmployeeID;
```

Employees - Outcome:

Employee	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	-	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

2. UNION AND UNION ALL

2.1

5. SELECT city AS location
FROM employees
UNION
SELECT status AS location projects;

Outcome:

Location
New York
Los Angeles
Toronto
London
Sydney
Completed
Ongoing
Pending
Pending

2.2

6. SELECT city AS location
FROM employees
UNION ALL
SELECT status AS Location
FROM Projects;

Outcome

Location
New York
Los Angeles
Toronto
London
Sydney
New York
Completed
Ongoing
Pending
Completed
Ongoing
Pending

3.

7. Filtering Statements

```
SELECT employeeID, FirstName, LastName,
       Department, salary
FROM employees
WHERE salary > 70 000;
```

Outcome:

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	clark	Finance	80000

~~3.2~~

8. SQL Query

```
SELECT employeeID, FirstName, LastName, Department,
       salary
FROM employee
WHERE Department IN ('IT', 'Finance');
```

Outcome:

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	clark	Finance	80000

9. Projects that are not completed

```
SELECT ProjectID; ProjectName, Budget, Status
FROM em Projects
WHERE status <> 'completed';
```

Outcome:

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	Ongoing
103	Cybesecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

10. Projects have a budget greater than 70000 and not completed

```
SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Budget > 70000 AND status <> 'completed';
```

Outcome:

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

11. Retrieve employees from New York OR Toronto, ordered by salary in descending order

SQL Query:

```
SELECT employeeID, firstName, LastName, Department,
       Salary, city
FROM employees
WHERE city ('New York', 'Toronto')
ORDERED BY salary DESC;
```

Outcome:

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

12. Top 3 highest-paid employees

SQL Query

```
SELECT employeeID, FirstName, LastName,
       Department, Salary
FROM Employees
ORDER BY Salary DESC,
LIMIT 3;
```

Outcome:

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	CLARK	Finance	80000
3	Bob	Johnson	Finance	75000
4	DAVID	Brown	IT	72000

4. Aggregate Functions with GROUP BY & HAVING

13.

Find the total salary per department in descending order

```
SELECT Department, SUM(Salary) AS totalSalary
FROM Employees
GROUP BY Department
ORDER BY Total salary DESC;
```

Outcome:

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

14.

Find average salary per city, where the average salary is greater than 65000.

```
SELECT city, AVG(salary) AS Average salary
FROM Employees
GROUP BY City
HAVING AVG(Salary) > 65000;
```

Outcome:

City	AverageSalary
Toronto	75000
New York	75000
London	72000

15.
Count the number of employees per departm-
ent, including only departments with more than
SQL Query

```
SELECT Department, COUNT(EmployeeID) AS
        EmployeeCount
FROM  Employees
GROUP BY Department
 HAVING COUNT (EmployeeID) > 1;
```

Outcome:

Department	EmployeeCount
IT	2
Finance	2

16.
Number of projects per status, but include
statuses with at least 2 Projects

SQL Query

```
SELECT status, ProjectCount(Rename COUNT
        (ProjectID) as ProjectCount);
FROM Projects
GROUP BY Status
 HAVING COUNT(ProjectID) >= 2;
```

Outcome:

STATUS	ProjectCOUNT
completed	2

17.
Total project budget per employee, only employees who are managing projects worth more than 150 000

```
SELECT employeeID, FirstName, LastName
SUM(Budget) AS TotalProjectBudget
FROM Employees e
INNER JOIN Projects p on e.EmployeeID = p.
        EmployeeID
GROUP BY e.EmployeeID, e.~~employee~~ ID
        FirstName, e.LastName
HAVING SUM(p.Budget) > 150 000;
```

Outcome

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000